

# Zhenhao Liu

Undergraduate Researcher in Natural Language Processing

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[lik1606.github.io](https://github.com/LikC1606) · [github.com/LikC1606](https://github.com/LikC1606) · [Westlake NLP Lab](#)

RESEARCH FOCUS NLP / Language Modeling / Cognitive Alignment / Research Agents

## EDUCATION

Westlake University

Undergraduate, Artificial Intelligence

Hangzhou, China

2024 cohort

## RESEARCH EXPERIENCE

Westlake NLP Lab

Hangzhou, China

[Advisor profile](#)

*Undergraduate Researcher; advised by Prof. Yue Zhang*

- Lead research on data-efficient Chinese language modeling and cognitive alignment, studying how representation depth and corpus-derived features affect word- and sentence-level fMRI encoding.
- Design reproducible analyses with controlled shared-representation comparisons, participant-cluster bootstrap, leave-one analyses, and sensitivity checks to separate task-specific effects from selection uncertainty.

## PUBLICATION

RepDuet at NLPCC 2026 Chinese BabyLM: Task-Specific Hybrid Representations for the Cognitive Track

[Model](#)

[Paper PDF](#)

*Sole first author; accepted to NLPCC 2026 Shared Tasks Poster*

- Developed RepDuet, a task-specific hybrid representation that combines intermediate contextual states with corpus statistics for word-level evaluation and upper-layer contextual states with lexical and confidence features for sentence-level evaluation.
- Ranked first in the eligible Cognitive Track after organizer reproduction and final-layer verification, achieving 0.33965 on the hidden evaluation; controlled development analyses found a positive margin over the strongest shared representation.

## RESEARCH INTERESTS

- **Cognitive NLP and language modeling:** data-efficient Chinese language models, representation learning, and neural encoding.
- **Research agents:** language-model agents for literature search, information synthesis, and research workflows.
- **AI for science:** AI systems that support scientific discovery and the everyday work of researchers.

## HONORS

|      |                                |              |
|------|--------------------------------|--------------|
| 2026 | ICPC Asia East Continent Final | Silver Medal |
| 2025 | ICPC Asia Xi'an Regional       | Gold Medal   |
| 2025 | CCPC Zhengzhou Site            | Gold Medal   |
| 2025 | ICPC Asia Shenyang Regional    | Silver Medal |

## TECHNICAL SKILLS

**Programming** Python, Git, command-line development, testing, and reproducibility

**AI/NLP** data-efficient language modeling, representation learning, cognitive NLP, LLM applications, language agents, evidence retrieval and synthesis, and AI for science

**Engineering** Flask, LangChain, REST APIs, Pytest, Ruff, Mypy, GitHub Actions, and LaTeX